

RISC and CISC are two different ways of designing computer processors.

- RISC uses a small set of simple, fixed-size instructions designed to execute in a single clock cycle.
- CISC includes a larger set of instructions, many of which are complex and can perform multiple operations (e.g., memory access and computation) in a single instruction.
- CISC instructions often require multiple clock cycles.

Reduced Instruction Set Computer (RISC)

- RISC stands for Reduced Instruction Set Computer. In Reduced Instruction Set Computer (RISC) architecture, the instruction set of the computer is simplified to reduce the execution time.
- RISC has a small set of instructions, which generally include register-to-register operations.
- Thus, data is stored in processor registers for computations, and results of the computations are transferred to the memory using store instructions.
- All operations are performed within the registers of the CPU. In RISC, all instructions have simple register addressing and hence use less number of addressing modes.
- RISC uses relatively a simple instruction format and is easy to decode. Here, the instruction length can be fixed and aligned on word boundaries.
- The RISC processors can execute one instruction per clock cycle.
- This is done using pipelining, which involves overlapping the fetch, decode, and execute phases of two or three instructions.
- As RISC takes relatively a large number of registers in the processor unit, it takes less time to execute its program when compared to CISC.
- RISC simplifies processor design by using a small, uniform set of instructions. Each instruction performs a basic operation (e.g., load, compute, store) and is designed to execute in a single clock cycle, enabling efficient pipelining and simpler hardware.

Characteristics of RISC

- Simpler instruction, hence simple instruction decoding.
- Instruction comes in the form of one word.
- An instruction takes a single clock cycle to get executed.

- More general-purpose registers for register-to-register operations.
- Simple Addressing Modes.
- Optimized for pipelining due to uniform instruction size and simplicity.

Complex Instruction Set Architecture (CISC)

CISC reduces the number of instructions a program needs by using a large set of complex, variable-length instructions. A single instruction can perform multiple operations (e.g., load, compute, and store), which may take multiple clock cycles.

Characteristics of CISC

- Complex instruction, hence complex instruction decoding.
- Instructions are larger than one-word size.
- Instruction may take more than a single clock cycle to get executed.
- Less number of general-purpose registers as operations get performed in memory itself.
- Complex Addressing Modes.

RISC	CISC
Small and simple instruction set	Large and complex instruction set
Fixed-length instructions	Variable-length instructions
Usually 1 cycle per instruction	Multiple cycles per instruction
More general-purpose registers	Fewer registers
Simple and limited addressing modes	Complex and many addressing modes
Larger code size (more instructions)	Smaller code size (fewer instructions)
Simple hardware, easier pipelining	Complex hardware, harder pipelining
Examples: ARM, RISC-V, MIPS	Examples: x86, Intel 80386